EXPERIMENTAL ANALYSIS OF TRANSPORT-LAYER DEMULTIPLEXING

THE PROBLEM

- Servers handle many connections simultaneously
- How are connections separated correctly?
- What happens under high load?

SYSTEM OVERVIEW

Clients → TCP Server → Dashboard

BACKEND ARCHITECTURE

- FastAPI backend
- TCP server (port 9000)
- Async load generator
- WebSocket updates

HOW CODE WORKS

- Async clients simulate connections
- Each client opens a TCP socket
- Server tracks connection ID and port

SCOPE OF THE PROJECT

What this project is

- Application-level analysis
- Real DNS lookups
- Performance & error behavior

What it is NOT

- Packet-level analysis
- DNS protocol internals

DEMULTIPLEXING

4-Tuple:

- Source IP
- Source Port
- Destination IP
- Destination Port

EXPERIMENT SETUP

- 10 clients
- 50 clients
- 100 clients
- Different request rates

RESULTS

- Active connections increased
- Message rate increased
- System remained functional

ANALYSIS

- Demultiplexing works correctly
- Performance changes under load
- Connection handling becomes complex

PORT ANALYSIS

- Unique source ports per connection
- Ephemeral port range

CONCLUSION

- Concept confirmed in practice
- Real system behavior observed

THANK YOU

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